User guide

Bow Tie Analysis – how to build a bow tie, manage the barriers that control a hazard, quantify the risk and find the barriers that matter most.

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1 · What a bow tie is

A **bow tie** shows on one page how a **hazard** can be released (the **top event** – the moment control is lost), what can cause it (**threats**, left), what can follow (**consequences**, right) and the **barriers** that stand in the way on each path – **preventive** on the left, **mitigative** on the right. **Escalation factors** are conditions that defeat or weaken a barrier; they have their own controls.

The left side is in effect a simplified fault tree (threats OR-ed into the top event, each blocked by barriers in series) and the right side a simplified event tree (barriers that must fail for each consequence). The bow tie is the communication and barrier-management view; the *Fault Tree* and *Event Tree* apps give the detail for selected barriers or outcomes.

2 · Quick start

1. Click the hazard box or the top event and describe both precisely.
2. + *Threat* / + *Consequence* in the side panel; give threats a frequency (/yr) and consequences a severity.
3. Select a threat or consequence and add barriers in the order they act (+ *New barrier*, or + *Existing barrier* to share one).
4. Click a barrier to set its type, PFD (or fault-tree link), owner, performance standard and verification; add escalation factors and their controls.
5. Read *Risk & barrier criticality*; switch to *Current barrier condition* in the header to see the effect of impaired barriers.

> The app opens with the hot DRI conveyor fire example (Section 13).

3 · Reading the diagram

- **Hazard** (striped box, top) → **top event** (orange circle, centre) with its frequency.
- **Threats** (blue, left) show *frequency* × *P(top | threat)* → *contribution to the top event*. **Consequences** (red, right) show residual frequency, severity S, likelihood L and risk rank.
- **Barriers** are coloured by type (legend) and show their PFD; **FT** = taken from a fault tree; ↺ = shared on several lines; **SCE** = marked safety-critical element. In the condition view a degraded barrier has an amber border, an unavailable one a red dashed border.
- **Escalation factors** (orange, dashed link) hang below the barrier they undermine, with their controls below them.
- Drag to pan, wheel or + / − to zoom, *Fit* to see the whole bow tie. Click any element to edit it.

4 · Hazard and top event

The **hazard** is something with the potential to cause harm that is part of normal business (e.g. “hot DRI on an enclosed belt conveyor”). The **top event** is the first moment control over that hazard is lost (e.g. “fire / smouldering of DRI on the belt”) – before any damage has occurred. Choose it so that threats lead into it and consequences follow from it; avoid making the top event a consequence (“conveyor destroyed”) or a cause (“belt stops”).

5 · Threats and preventive barriers

- A **threat** is a credible cause that can, on its own, release the hazard. Give each a **frequency** (/yr) and, where useful, **P(top | threat)** – the probability that the threat leads to the top event when every preventive barrier fails (e.g. the chance that stationary DRI actually ignites). Leave 1 if every occurrence leads to the top event.
- **Preventive barriers** stop the threat developing into the top event. Each must, on its own, be able to stop that threat (effective, independent, auditable). Order them left to right as they act.

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contribution of a threat = frequency × P(top | threat) × Π PFD(preventive barriers on that line)
top event frequency = Σ contributions of all threats
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6 · Consequences and mitigative barriers

- A **consequence** is an outcome of the top event. Rate its **severity** 1–5 for people, environment, asset / production and reputation; the highest rating is used.
- **P(consequence | top)** is the probability that the top event leads to this consequence if every mitigative barrier on the line fails (e.g. the probability that people are present). Leave 1 if it always would.
- **Mitigative barriers** reduce the chance or the severity of the consequence once the top event has happened (detection, suppression, containment, emergency response).

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residual frequency = top event frequency × P(cons | top) × Π PFD(mitigative barriers on that line)
inherent frequency = Σ(threat frequency × P(top | threat)) × P(cons | top) (no barriers at all)
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7 · Barriers – types, properties, sharing

Type	Meaning	Example
Passive hardware	works by its presence, no activation	fire walls, bunds, fire separation of a gallery
Active hardware	detect – decide – act, all automatic	trips, N2 inerting system, sprinklers
Active human-hardware	a person acts on an instrument / device	operator responds to an alarm, manual fire fighting
Active human	a person detects, decides and acts	permit-to-work checks, cleaning before hot work
Continuous / procedural	ongoing activity or management system	housekeeping, condition monitoring, inspection

Properties recorded in the barrier register: PFD (design), status (Available / Degraded / Unavailable) and degradation factor, safety-critical element (SCE), **owner**, **performance standard** (what the barrier must achieve – functionality, availability, response time) and **verification** (how it is tested or assured). These turn the bow tie into a barrier-management tool.

Shared barriers: use + *Existing barrier* to put the same barrier on several lines (e.g. the N2 system on three consequences). It is one register entry – edits and status changes apply everywhere. A barrier used on both the prevention and mitigation sides is flagged, because a single failure would defeat both.

8 · Escalation factors and their controls

An **escalation factor** is a condition that makes a barrier fail or less effective (e.g. “low N2 buffer pressure” for the N2 system, “power failure” for an emergency restart). **Escalation-factor controls** keep the barrier effective (e.g. low-pressure alarm with operator response, locked-open isolation valve). Select a barrier and use + *Escalation factor*; select the factor and use + *Control*.

Escalation factors are qualitative in this app – their effect is reflected by the barrier PFD (or by the fault tree, where the escalation factors usually appear as basic events) and by the barrier status.

9 · Quantification and the risk matrix

Each consequence is placed in a 5 × 5 matrix: **severity** S (highest of the four dimensions) × **likelihood class** L from its frequency, using the bands on the *Risk matrix* tab (default: L1 < 10^{-4}, L2 10^{-4}–10^{-3}, L3 10^{-3}–10^{-2}, L4 10^{-2}–10^{-1}, L5 ≥ 10^{-1} per year). Risk score = S × L; rank Low < 5 ≤ Medium < 10 ≤ High < 20 ≤ Very High (editable).

Optionally enter a **tolerable frequency** per severity level; consequences above it are flagged with the exceedance factor.

The *Risk & barrier criticality* tab shows inherent risk (no barriers) and residual risk (with barriers) for every consequence.

10 · Barrier condition (health) view

The header button switches between **Design view** (all barriers at design PFD) and **Current barrier condition**. In the condition view a *Degraded* barrier's PFD is multiplied by its degradation factor (default 10, capped at 1) and an *Unavailable* barrier has PFD = 1. Use it with live barrier status (inspection findings, bypasses, overdue tests) to see the risk the plant is actually running and to support "barrier impairment" decisions.

11 · Barrier criticality and safety-critical elements

For every barrier the app sets its PFD to 1 (barrier failed) and recalculates every consequence. The table shows the resulting frequencies and risk ranks and the largest increase factor. A barrier is **suggested as safety-critical** if its failure would raise a consequence into a higher High / Very High rank, or increase a consequence frequency 10-fold or more. Confirm with the team and mark it as SCE in the register – SCEs need a performance standard, an owner and a verification scheme.

12 · Linking barriers to fault trees

Import a fault tree (*File ▸ Import fault tree*) from the **Fault Tree & PFD Calculator**, then on the barrier set *PFD source* to *fault tree gate* and choose the gate that represents "barrier fails on demand" – for a protection system the gate below any initiating event (not a frequency). The fault tree is stored in the bow-tie file and evaluated with the same engine (exact, time-averaged PFD). Replace it with a newer version on the *Fault tree links* tab after changes.

13 · Worked example – hot DRI conveyor fire

Hazard: hot DRI on an enclosed belt conveyor. **Top event:** fire / smouldering of hot DRI on the belt. All data are illustrative.

Threat	f /yr	P(top|threat)	Preventive barriers (PFD)	To top /yr
Trip / stoppage under load	6	0.2	restart / emergency discharge (0.25)	0.30
Over-temperature DRI discharged	0.5	0.3	high-temperature trip (0.01), operator response (0.1)	1.5×10^{-4}
Seized idler / friction hot spot	0.2	0.2	condition monitoring (0.3), belt-sway / speed trip (0.1)	1.2×10^{-3}
Hot work near the conveyor	2	0.05	permit-to-work (0.1), fines cleaning (0.3)	3×10^{-3}
Fines accumulation	1	0.1	housekeeping (0.3), dust extraction (0.1)	3×10^{-3}

Top event ≈ **0.31 /yr**, almost entirely from trips under load – the same 0.3 /yr fire frequency as the event tree example (6 × 0.25 × 0.2).

Consequence	S	Mitigative barriers (PFD)	Residual /yr	Risk
Major belt fire – extended outage	4	N2 system (0.0285, from fault tree G-002), manual fire fighting (0.2)	1.8×10^{-3}	High (L3)
Fire spreads to transfer tower / silo	5	N2, manual fire fighting, fire separation (0.1)	1.8×10^{-4}	High (L2) – above the illustrative 10^{-4} /yr
Personnel exposed to smoke / CO (P = 0.1 people present)	4	CO detection & evacuation (0.1), restricted access (0.1)	3.1×10^{-4}	Medium
Asphyxiation after N2 release (P = 0.05 entry)	5	N2 release warning & lock-out (0.1), O2 monitoring & entry permit (0.1)	1.5×10^{-4}	High – above 10^{-4} /yr

Lessons: (1) the N2 system is shared by the two fire consequences and its failure would raise the major-fire frequency about 35-fold – it is the key safety-critical element, with escalation factors (low N2 pressure, open enclosure doors) that need controls; (2) the barrier itself introduces a new hazard – **N2 asphyxiation** – which needs its own barriers; (3) the dominant threat is the trip under load, so reducing stoppages or improving the restart / emergency discharge (and its power-failure escalation factor) reduces every consequence. Try: set the N2 barrier to *Unavailable* and switch to *Current barrier condition*.

14 · Saving and export

- Automatic browser storage; *File ▸ Save as…* (Chrome / Edge) links a `.bowtie.json` file that is written after every change; *Open file…*; *Download / Import backup*.
- *Export register & results to Excel*: summary, barrier register, threats, consequences, escalation factors, criticality.

- *Export bow tie picture (.svg)* for reports and posters; *Print report / PDF* (A3 landscape).

15 · Good practice and limitations

Good practice

- One hazard and one top event per bow tie; keep threats and consequences specific.
- Only credit a barrier if it is effective on its own, independent of the threat and of the other barriers on the line, and auditable; activities that support barriers (training, maintenance) belong as escalation-factor controls, not as barriers.
- Give every safety-critical barrier an owner, a performance standard and a verification activity.
- Review the bow tie with operations and maintenance; keep barrier status current.

Limitations

- Barriers on a line are treated as independent and in series; threats are summed (rare-event approximation); a shared barrier is not double-credited on a line but dependencies between different barriers are not modelled – use the fault tree for that.
- Escalation factors are qualitative.
- Frequencies and PFDs are point values; results are for analysis, training and barrier management, not a substitute for a full QRA or SIL verification.

16 · FAQ

Why is the residual risk still High although there are barriers?

Severity 5 consequences need a very low frequency to leave the High band. Add or improve barriers, reduce the threat frequency, or check that the tolerable frequency is the right criterion.

“Barrier used on both sides”.

The same barrier cannot independently prevent and mitigate; credit it on one side, or split it into two distinct functions.

“Linked fault-tree gate not found / is a frequency”.

Re-select the gate on the barrier; use the protection-system gate below any initiating event.

Deleting a barrier.

Remove it from each line; it disappears from the register when it is no longer used anywhere (its escalation factors go with it).

17 · Glossary and references

Hazard	Source of potential harm that is part of the business.
Top event	Release of the hazard – loss of control.
Threat	Possible cause of the top event.
Consequence	Possible outcome of the top event.
Barrier	Control that prevents a threat from causing the top event (preventive) or the top event from leading to a consequence (mitigative).
Escalation factor	Condition that defeats or degrades a barrier; controlled by escalation-factor controls.
PFD	Probability of failure on demand of a barrier.
SCE	Safety-critical element – a barrier whose failure could cause or contribute substantially to a major accident.

- CCPS & Energy Institute (2018) – Bow Ties in Risk Management: A Concept Book for Process Safety.
- CCPS – Layer of Protection Analysis: Simplified Process Risk Assessment; Guidelines for Initiating Events and Independent Protection Layers in LOPA.
- ISO 31010:2019 – Risk management – Risk assessment techniques (bow-tie analysis).
- IOGP Report 415 – Asset integrity: the key to managing major incident risks.